

Spatial Blocking Does Not Prevent Group-Aggregate Leakage: A Case Study in Urban Turnover Prediction

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Abstract

Two evaluation hazards are individually well documented: spatially blocked cross-validation is required for geographic data, with block size the dominant choice [Roberts et al., 2017, Ploton et al., 2020, Stock, 2025], and target-encoded categorical features leak unless folds respect the encoding groups [Micci-Barreca, 2001, Prokhorenkova et al., 2018]. We show that these interact in a way that defeats the standard remedy, and that the interaction is easy to miss because it produces no visible symptom.

On a task built from 1.21 million City of Chicago business licences spanning 1995–2026, we predict which of 135,553 storefronts exhibit excess tenant turnover. Random k -fold cross-validation reports ROC AUC 0.970 and average precision 0.756 ($15.1\times$ the base rate); partitioning by administrative unit reports 0.693 and 0.114 ($2.3\times$).

Our contribution is to locate the channel. Ablation shows the inflation is carried almost entirely by group aggregates—ward and community-area leave-one-out means, which inflate AUC by 0.321—rather than by proximity: neighbourhood features computed at 50–500 m inflate by only 0.034. Because the channel is group membership rather than distance, spatial blocking does not close it at any scale a practitioner would plausibly select. Blocks of 100 m to 1 km leave performance statistically unchanged (AUC 0.971–0.977 against 0.970 for random splits); only blocks approaching the size of the administrative units themselves recover the honest estimate. A practitioner following current guidance—choosing block size from the autocorrelation range of the predictors—would therefore still obtain an inflated estimate here, because the relevant scale is set by the aggregation units, not by the autocorrelation of the data.

A controlled experiment isolates the mechanism. Varying only the size of the encoding group, a single leave-one-out group-mean feature reaches AUC 0.986 under random folds while its group-blocked value stays near chance, and inflation falls from $+0.415$ to $+0.088$ as group size falls from 13,555 to 33. Inflation tracks group size rather than how much groups differ, as the algebra requires: leave-one-out withholds a $1/|G|$ share of the signal, which is negligible for administrative units. We further confirm in this domain that leakage equalises model families under honest evaluation, inflation ranging from 0.013 AUC for logistic regression to 0.277 for gradient boosting, so random splits corrupt model selection as well as assessment.

The substantive finding survives honest evaluation and is, to our knowledge, new: excess turnover concentrates at individual addresses rather than districts, with neighbours only weakly elevated and tenancies failing roughly $1.5\times$ faster per renewal cycle. The underlying records are public domain, and the pipeline is available from the authors on request.

*Code and derived data are available from the authors on request.

1 Introduction

Applied machine learning on geographic data has a well-documented evaluation problem. Observations close in space share unobserved causes, so a randomly partitioned test set is not independent of the training set and performance estimates are optimistic. The remedy—partitioning by spatial block rather than at random—is established [Roberts et al., 2017, Ploton et al., 2020, Valavi et al., 2019], and recent work identifies block size as the choice that matters most, recommending it be set from the autocorrelation range of the predictors [Stock, 2025].

Separately, it is well known that target-encoded categorical features leak: encoding a category by its mean outcome transmits label information, and the standard precautions are to compute the encoding within training folds and to partition by group so that a category cannot span the split [Micci-Barreca, 2001, Prokhorenkova et al., 2018]. Modern tooling supports this directly.

This paper concerns what happens when both apply at once, which is the normal situation in urban prediction. Datasets built from civic records routinely carry both geographic coordinates and aggregates over administrative units—ward means, district rates, neighbourhood statistics—and the two hazards are commonly treated as one. We find they are not, and that treating them as one leaves the practitioner with an inflated estimate and no indication of it.

Contribution. We do not claim that spatial leakage, block-size sensitivity or target-encoding leakage are new; each is established in the work cited above. Our contributions are as follows.

1. **The channel is group membership, not proximity.** Ablation on a real task attributes the inflation almost entirely to administrative aggregates (+0.321 AUC) rather than to distance-based neighbourhood features (+0.034). The leak is therefore not spatial autocorrelation in the sense the spatial cross-validation literature addresses.
2. **Spatial blocking does not close it.** Because the channel is group membership, blocks must reach the size of the aggregation units before the estimate becomes honest. Blocks from 100 m to 1 km—the range a practitioner would select from predictor autocorrelation following Stock [2025]—leave performance statistically unchanged. Existing block-size guidance is thus insufficient when group aggregates are present, because the governing scale is set by the aggregation units rather than by the autocorrelation of the data.
3. **The mechanism, isolated and quantified.** In a controlled experiment varying only encoding-group size, a single group-mean feature reaches AUC 0.986 under random folds against near-chance when folds are group-aligned, and inflation scales with group size rather than with between-group variance. Leave-one-out withholds a $1/|G|$ share of the signal, so it is not a meaningful precaution for groups the size of administrative units. This accounts quantitatively for the ablation above.
4. **A quantification and a diagnostic in a new domain.** We report the magnitude on 1.21M records spanning thirty years, where average precision inflates more than sixfold, and give a cheap detector: sweep block size, and a flat curve indicates blocks below the governing scale. We also confirm in this domain the finding of Rosenblatt et al. [2024] that model families differing sharply under leakage become equivalent once it is removed. The evaluation results are invariant to the target definition, reproducing to within 0.045 AUC across three definitions that select substantially different locations.
5. **A substantive result.** Excess commercial turnover concentrates at individual addresses rather than districts. We are not aware of prior location-level work of this kind; the firm-exit

literature operates at firm and sector level, and municipal licence registers remain little used despite recording entry and exit directly.

2 Related work

Spatial cross-validation. Roberts et al. [2017] give the standard treatment of blocked designs for structured dependence, and Ploton et al. [2020] show that ignoring spatial autocorrelation inflates reported accuracy. Valavi et al. [2019] provide tooling that estimates the autocorrelation range of predictors in order to set block size. Stock [2025] study the choice systematically across 1,426 synthetic datasets and conclude that block size dominates block shape, fold count and fold assignment, with predictor correlograms a practical guide.

Our results are consistent with that work and extend it in one respect. Its guidance sets block size from the spatial structure of the data. Where features include aggregates over administrative units, we find the governing scale is instead the size of those units, which may exceed the autocorrelation range and is not recoverable from a correlogram. Applying the existing guidance to our task yields blocks that leave the estimate fully inflated.

Leakage and target encoding. Kaufman et al. [2012] give the canonical taxonomy and Kapoor and Narayanan [2023] document prevalence across fields. Micci-Barreca [2001] introduced target encoding and Prokhorenkova et al. [2018] analyse its leakage, the accepted remedies being to compute encodings inside training folds and to partition by group. Cerqua et al. [2024] identify spatial leakage as a form of cross-sectional leakage with aggregate panel data. Our contribution is not to restate these results but to show that in a geographic setting the group remedy and the spatial remedy are not interchangeable: practitioners reach for the spatial one, and here it does not work.

Leakage and model comparison. Rosenblatt et al. [2024] show in neuroimaging that flexible models lose their apparent advantage over logistic regression once leakage is corrected. We observe the same pattern and report it as a replication in a new domain rather than as a new finding.

Business survival. Firm exit has a long empirical literature [Dunne et al., 1988], typically at firm or sector level. Work on location-level persistence is comparatively sparse, and municipal licence registers remain an under-used source despite recording entry and exit dates directly.

3 Data and task

3.1 Source

The City of Chicago publishes every business licence issued since 1995 as public domain data.¹ We use the complete extract of 1,207,055 licence records retrieved 2026-09-18.

We note that the portal’s one-click CSV export returns approximately 228k rows with HTTP 200 and no warning. All results here use the paginated API with the row count verified against the server’s own total. We mention this because a truncated extract produces plausible output and is difficult to detect downstream.

¹Dataset `r5kz-chrr`, <https://data.cityofchicago.org>.

3.2 From licences to tenancies

A licence is not a storefront, and three properties of the register would inflate apparent turnover if taken at face value. Licences renew on one- or two-year terms, so a continuously trading business appears as many rows. Businesses hold several licence classes concurrently (a restaurant may hold food, liquor and sidewalk-cafe licences). Ownership restructuring or renaming ends one licence and begins another while trading is uninterrupted.

We therefore collapse licences into tenancies in two stages. Licences sharing an account number at one address are merged into a single occupancy when consecutive spells are separated by less than 200 days, absorbing renewals and concurrent classes. Occupancies by different account holders at one address are then merged when separated by less than 30 days, on the grounds that a handover with no vacancy is more likely a restructuring than a failure and replacement. The succession threshold is deliberately short, since aggressive merging suppresses the signal being measured.

Addresses are normalised before grouping (suffix and directional canonicalisation, ordinal collapse, accent stripping), with unit designators retained so that separate suites in one building remain separate storefronts.

3.3 Censoring and quantization

Two measurement properties constrain what may be inferred.

Licence terms extend into the future—this extract contains expiry dates as late as 2030. Anchoring the observation point to the latest date present would place it years after the data ends, and every currently trading business would appear as a completed tenancy credited with unelapsed time. We anchor to the latest issue date (2026-09-18); 38,143 of 194,203 tenancies are right-censored.

Tenancy duration is quantized to licence terms. Only 6.2% of tenancies end with a recorded cancellation and therefore an exact date; the remaining 93.8% lapse, and are known only to have ended within a term. Observed durations concentrate on exact multiples of licence terms (729–730 days, 364–365, 1460). Duration in days is consequently closer to a statement about the municipal renewal calendar than about trading life, and we use renewal cycles as the time axis for survival analysis.

3.4 Target definition

Raw tenant count conflates three things that are not properties of the address: exposure (an address first licensed in 1995 has had thirty years to turn over), sector base rate (hospitality fails more often than professional services), and district (some corridors churn throughout).

We therefore fit expected turnover with a negative-binomial GLM using $\log(\text{exposure})$ as offset and dominant licence class and community area as covariates, and define *excess turnover* as the residual. Locations in the upper 5% of excess are labelled positive. A location is not positive for turning over often, but for turning over more than its own context predicts.

Restricting to locations with at least eight years of exposure leaves 135,553 locations across 78 community areas, of which 6,778 are positive. Positive locations average 3.12 tenants against 1.13 for the remainder.

We note that community area enters the target definition. Comparisons of positives against negatives at community-area level are therefore partly circular, and we exclude them from substantive claims; the 50–500 m and building-level comparisons are out-of-model and are used instead.

Partition	Blocks	ROC AUC	AP	Lift
Random (none)	—	0.970 ± 0.004	0.756	$15.1\times$
Grid 100 m	20,822	0.977 ± 0.003	0.755	$15.1\times$
Grid 250 m	7,111	0.977 ± 0.002	0.748	$15.0\times$
Grid 500 m	2,202	0.976 ± 0.002	0.760	$15.2\times$
Grid 1 km	632	0.971 ± 0.005	0.731	$14.6\times$
Grid 2 km	189	0.942 ± 0.014	0.618	$12.4\times$
Grid 5 km	40	0.825 ± 0.091	0.208	$4.2\times$
Community area	78	0.693 ± 0.062	0.114	$2.3\times$

Table 1: Performance against partitioning scheme. Data, features and model are identical throughout; only the partition changes. Blocking at 100 m–1 km is indistinguishable from no blocking.

3.5 Features

All contextual features are computed leave-one-out, excluding the focal location. A neighbourhood statistic that includes the focal location partially reproduces the label.

Features comprise: leave-one-out mean turnover among neighbours within 50, 200 and 500 m and neighbour counts at each radius; leave-one-out means over ward and community area; building-level context (units in building, leave-one-out turnover among other units); and coordinates. Exposure terms are excluded from predictive experiments because exposure enters the target definition.

4 Experimental protocol

We compare partitioning schemes holding data, features and model fixed.

Random. Stratified 5-fold, shuffled—the default in common tooling.

Spatial grid. Space is tiled into squares of edge $s \in \{100, 250, 500, 1000, 2000, 5000\}$ m; whole squares are assigned to folds, so locations within a square cannot straddle the split.

Administrative. Whole community areas (78) are assigned to folds, balanced greedily by size.

We report ROC AUC (mean and standard deviation across folds) and average precision computed on pooled out-of-fold predictions. Average precision is reported because the positive rate is 5%, at which AUC is a flattering summary; the base rate gives a directly interpretable reference.

For the substantive spatial contrast we use a permutation null that shuffles labels *within* community areas, preserving each district’s turnover level so that only the assignment of labels inside it is randomised. A citywide shuffle would destroy district-level clustering and render almost any contrast significant.

5 Results

5.1 Block size must exceed feature support

Table 1 and Figure 1 show performance is flat from random splits through 1 km blocks and falls only beyond 2 km. Since features summarise neighbourhoods to 500 m, blocks at or below that scale cannot sever the dependence they induce: information propagates as far as the blocks are wide.

The practical implication is a diagnostic rather than a fixed recommendation. Sweeping block size and observing a flat curve indicates blocks are too small relative to feature support; the curve’s decline locates the scale at which dependence is actually broken.

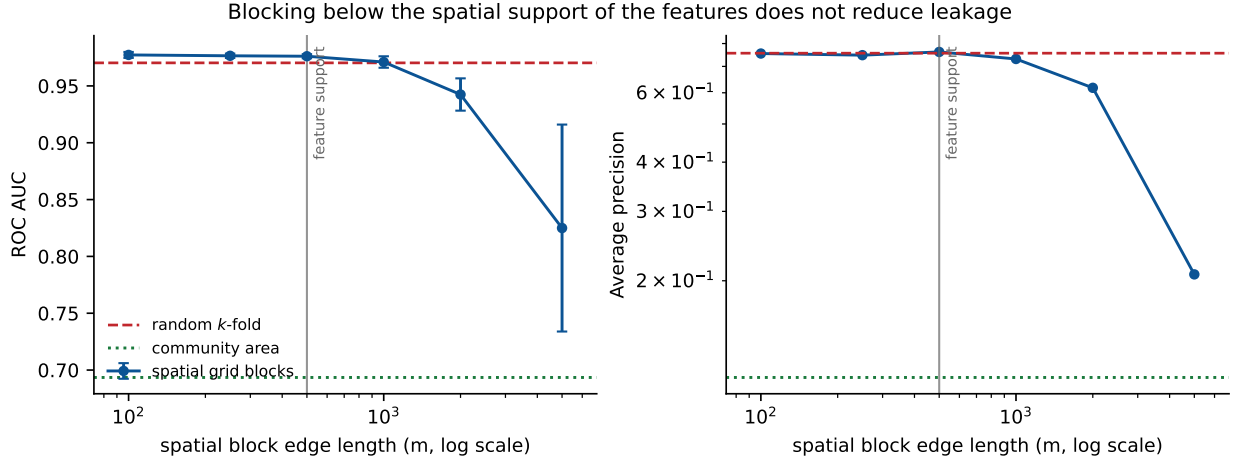


Figure 1: Performance against spatial block size. The curve is flat until blocks exceed the 500 m spatial support of the features (grey line), then falls. A practitioner selecting a block below that scale obtains an estimate indistinguishable from random splits.

Model	Random AUC	Blocked AUC	Δ	Random AP	Blocked AP
Logistic regression	0.709	0.696	+0.013	0.098	0.098
k NN ($k = 25$)	0.686	0.616	+0.070	0.092	0.072
Random forest	0.933	0.686	+0.247	0.670	0.086
Gradient boosting	0.970	0.693	+0.277	0.756	0.114

Table 2: Inflation by model family. Under blocked evaluation all families are equivalent; under random splits they are not.

5.2 Leakage scales with model capacity

Table 2 and Figure 2 show inflation rising with model flexibility. The consequence extends beyond reporting. Under random cross-validation gradient boosting appears to outperform logistic regression by 0.261 AUC; under blocked evaluation the gap is -0.003 . Random splits therefore do not merely inflate a figure, they invert a model-selection decision, and the inflation is largest precisely for the model families practitioners most often prefer.

5.3 Group aggregates dominate the leak

Table 3 and Figure 3 localise the leak to administrative aggregates. Four features alone reach AUC 0.978 under random splits and 0.657 under blocked splits. Proximity features—the expected culprit—contribute little.

The mechanism is group identity rather than distance. A leave-one-out mean over a ward is near-constant within that ward, so it serves as a ward identifier; under a random split, training rows disclose the ward’s label rate to test rows in the same ward. Leave-one-out construction removes the focal observation but not the remaining $n - 1$, and with large groups this is negligible protection.

This reframes the requirement. What must align is not blocks with geography but *the evaluation partition with the grouping used to construct aggregate features*. Where those groups are

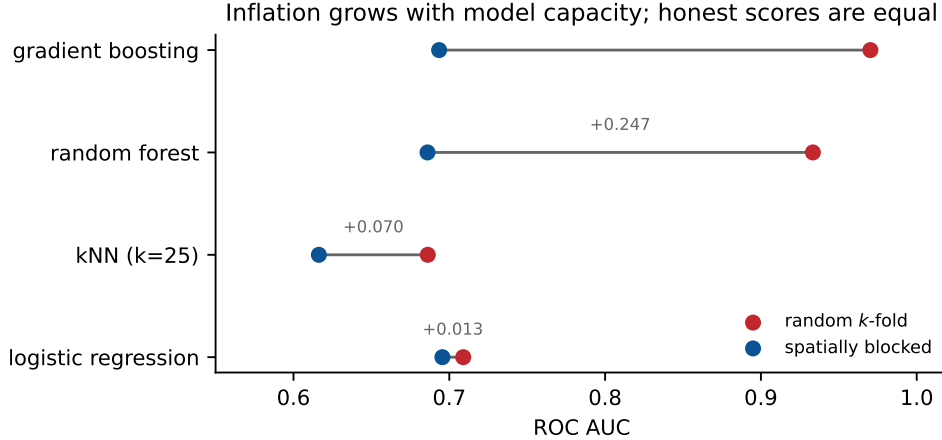


Figure 2: Paired random and blocked scores by model family. Honest scores coincide; random-split scores do not, so the ranking is an artefact of the partition.

Feature family	k	Random	Blocked	Δ
Area aggregates (ward, community area)	4	0.978	0.657	+0.321
Raw coordinates	2	0.659	0.595	+0.064
Neighbourhood LOO (50–500 m)	6	0.722	0.687	+0.034
Density counts	3	0.694	0.672	+0.021
Building context	2	0.508	0.506	+0.001

Table 3: Inflation by feature family, gradient boosting. Administrative aggregates inflate an order of magnitude more than proximity features.

administrative units, spatial blocks of arbitrary size do not satisfy it. Where the groups are not spatial at all—user, merchant, session—the same failure arises with no spatial character, and the same remedy applies.

5.4 Isolating the mechanism

The ablation identifies *which* features leak but not *why*, and on observational data the two are hard to separate: administrative aggregates differ from neighbourhood features in group size, in spatial extent and in how strongly their groups differ from one another. We therefore construct encoding groups directly and vary one property at a time.

Locations are partitioned into K spatial clusters for K between 10 and 4,000, giving mean group sizes from 13,555 down to 33. For each K we build a single feature—the leave-one-out group mean of tenant count—and evaluate it alone, under random folds and under folds aligned with the clusters.

Three things follow from Table 4.

First, the magnitudes are not small. *One* feature reaches AUC 0.986 under random folds. Nothing else is supplied to the model.

Second, the group-blocked column stays between 0.487 and 0.611 throughout. The feature carries almost no signal that transfers to unseen groups, so essentially the whole random-fold score is leakage rather than a genuine relationship that blocking happens to suppress.

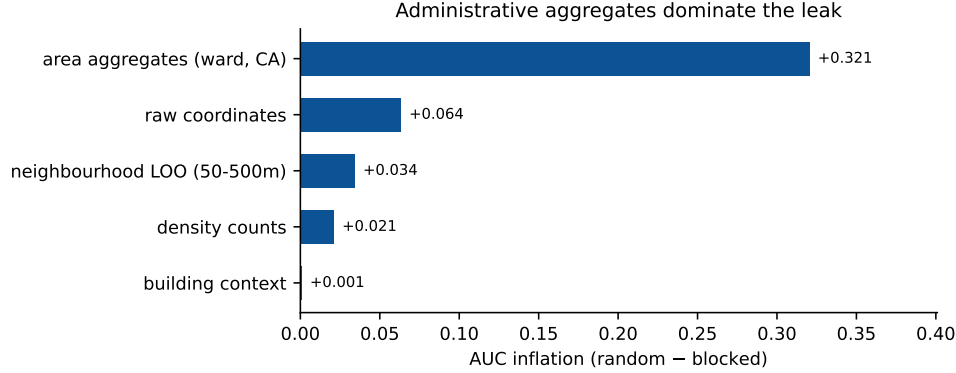


Figure 3: AUC inflation by feature family. Administrative aggregates dominate; proximity features contribute little.

Groups	Mean size	Between-group var.	Random	Group-blocked	Δ
10	13,555	0.00044	0.986	0.571	+0.415
25	5,422	0.00036	0.987	0.562	+0.424
50	2,711	0.00032	0.943	0.539	+0.404
100	1,355	0.00032	0.869	0.487	+0.382
250	542	0.00051	0.803	0.549	+0.254
1,000	135	0.00121	0.720	0.563	+0.157
4,000	33	0.00364	0.699	0.611	+0.088

Table 4: A single leave-one-out group-mean feature, evaluated under random and group-aligned folds, as the size of the encoding group varies.

Third, inflation tracks group size and not how much the groups differ. Between-group variance in the label rate rises by a factor of eight as groups shrink, while inflation falls from +0.415 to +0.088. Were leakage driven by groups differing from one another, the two columns would move together; they move oppositely.

The algebra gives the reason. For a group G the leave-one-out encoding is

$$f_i = \frac{1}{|G| - 1} \sum_{j \in G, j \neq i} y_j \quad \text{so} \quad f_i = \bar{y}_G + \frac{\bar{y}_G - y_i}{|G| - 1}.$$

The encoding differs from the full group mean—the quantity that would leak y_i outright—by a term of order $1/|G|$. Leave-one-out therefore withholds a $1/|G|$ share of the signal and no more: about 3% at $|G| = 33$, and 0.007% at $|G| = 13,555$. For large groups it is not a meaningful precaution, and what remains is a near-constant value within each group, which under random folds serves as a group identifier whose label rate the training rows disclose.

This accounts for the ablation in Table 3 quantitatively. Chicago’s 50 wards and 78 community areas hold on the order of 1,700 locations each, placing them in the regime where leave-one-out protects almost nothing; they inflate by +0.321. Neighbourhood features at 50–500 m aggregate over a handful of premises, where the $1/|G|$ term is substantial; they inflate by +0.034. The distinction is group size, not distance, which is why partitioning on geometry does not address it.

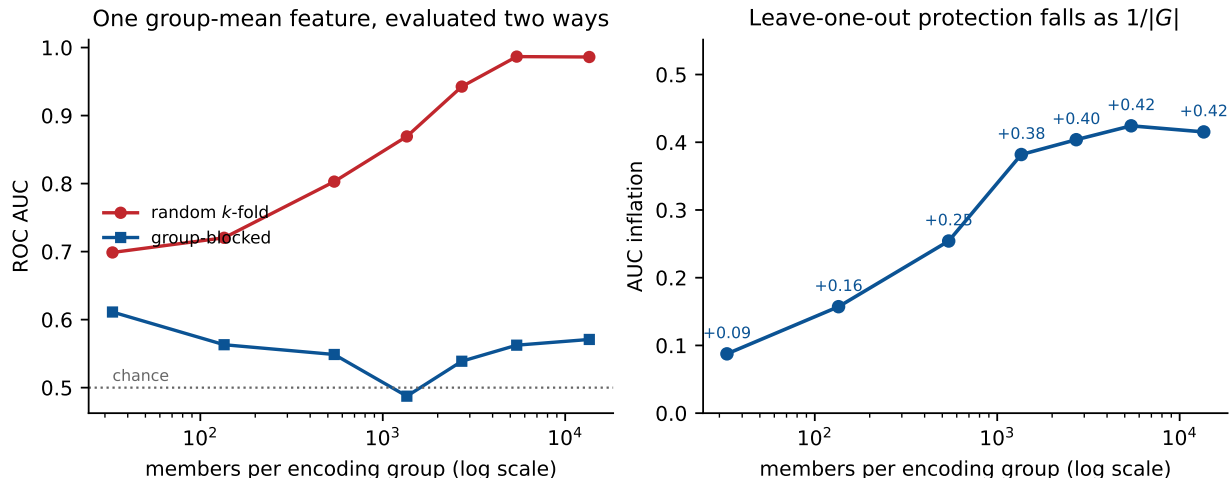


Figure 4: Leakage against encoding-group size. One feature reaches AUC 0.986 under random folds while its group-blocked value stays near chance.

	Count	Rate	Excess
Count (≥ 4 tenants)	1.000	0.187	0.193
Rate (tenants per decade, top 5%)	0.187	1.000	0.632
Excess (model residual, top 5%)	0.193	0.632	1.000

Table 5: Jaccard overlap between the sets of locations flagged by three target definitions.

5.5 The substantive finding

Under honest evaluation the urban result stands. Excess turnover concentrates at individual addresses rather than districts. Leave-one-out neighbour turnover exceeds the within-area permutation null by +0.104 tenants at 50 m ($p = 0.001$, null $+0.034 \pm 0.003$), +0.068 at 200 m and +0.040 at 500 m, against a focal difference of +1.99 tenants. Neighbours of high-turnover addresses are elevated by roughly 5%; the addresses themselves by roughly 176%.

Discrete-time survival over renewal cycles (Figure 6) shows tenancies at these locations ending faster at every stage: first-cycle hazard 0.337 against 0.233, and survival to the eighth cycle of 0.119 against 0.278. The hazard ratio is approximately constant across cycles, consistent with proportional hazards.

Context predicts the label only weakly (AUC 0.693, AP $2.3\times$ base rate). Most excess turnover is therefore not explained by observable surroundings.

5.6 Sensitivity to the target definition

Table 5 reports agreement between the three definitions. The exposure-adjusted definitions overlap substantially (Jaccard 0.632), but the unadjusted count-based definition overlaps both only weakly (0.187, 0.193). This is a genuine limitation and we state it plainly: which addresses are called high-turnover depends materially on whether exposure and sector are adjusted for, and a reader who prefers a raw-count definition would be studying a substantially different set of locations.

This sensitivity concerns the substantive finding rather than the evaluation findings, and we verify rather than assume the distinction. Repeating the random-versus-blocked comparison under

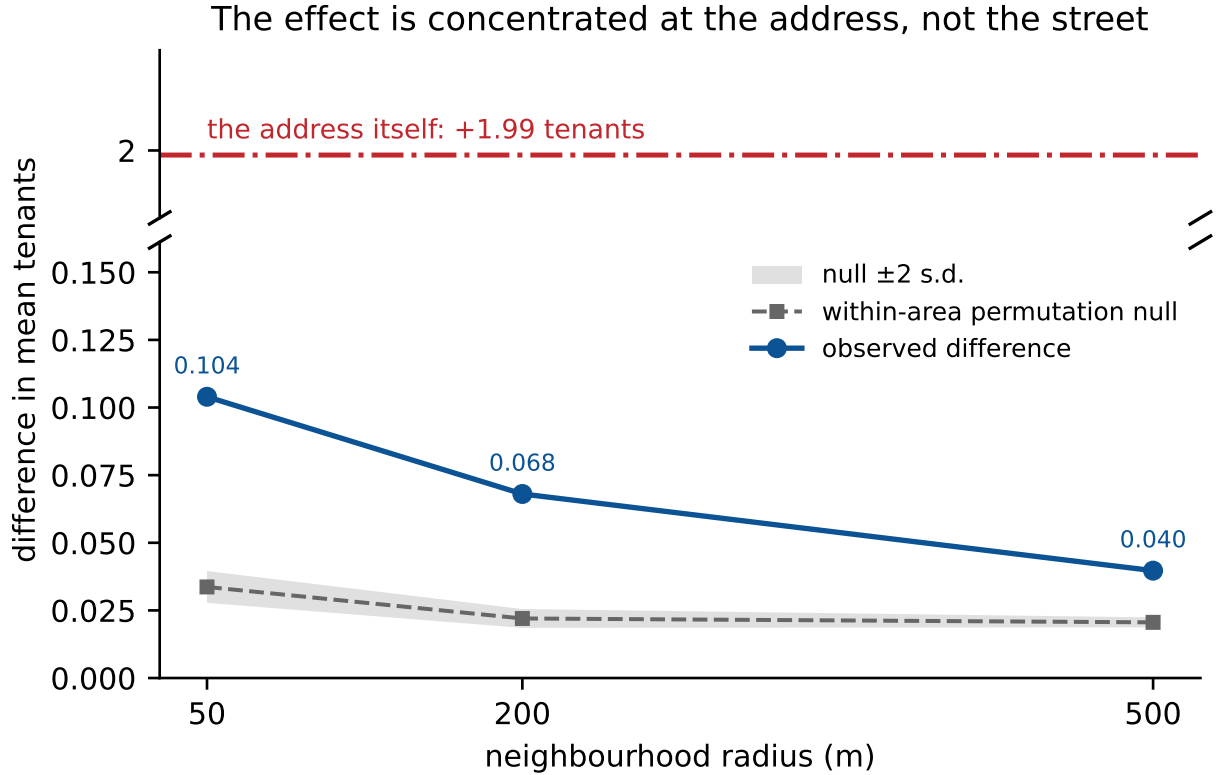


Figure 5: Difference in mean tenant count between high-turnover locations and the remainder, by neighbourhood radius, against a permutation null that shuffles labels within community areas. The focal difference exceeds the neighbour difference by an order of magnitude.

each definition gives inflations of +0.340 (count), +0.305 (rate) and +0.295 (excess), with random-fold AUC between 0.951 and 0.977 and blocked AUC between 0.637 and 0.676. The three definitions select substantially different locations, yet the leakage they exhibit is the same to within 0.045 AUC. The evaluation findings are therefore invariant to a choice that materially changes the substantive one, as the mechanism implies: leakage depends on how the features are constructed, not on how the label is defined.

6 Recommendations

1. **Partition by the grouping used for aggregate features.** If features include group-level statistics, folds must respect those groups. This subsumes spatial blocking when groups are geographic and applies unchanged when they are not.
2. **Report a block-size sweep.** A flat performance curve across block sizes indicates blocks below feature support. The curve costs little and makes the claim auditable.
3. **Treat leave-one-out as insufficient.** It removes the focal observation only. With large groups the remaining members carry essentially the full signal.
4. **Compare model families under the final protocol.** Capacity-dependent inflation means selection under random splits can invert under blocked splits.

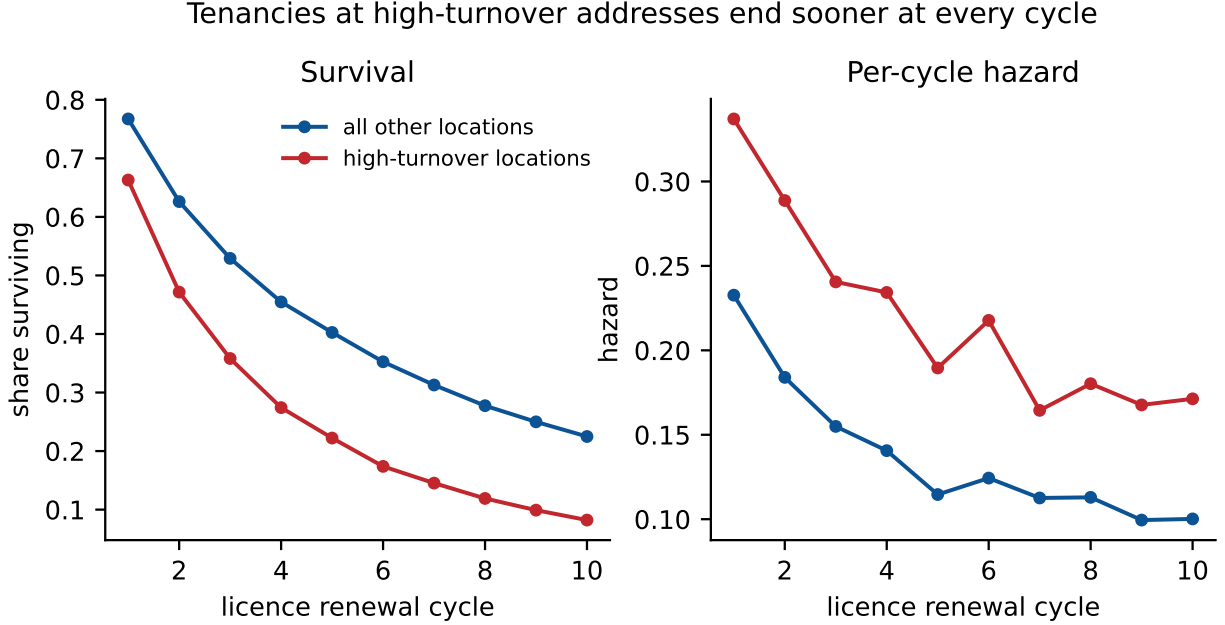


Figure 6: Discrete-time survival and per-cycle hazard over licence renewal cycles. Renewal cycles are used in place of elapsed days because duration is quantized to licence terms.

5. **Report average precision with the base rate** for imbalanced targets.

7 Limitations

Licence records proxy trading. A lapse may reflect administrative change rather than closure; our succession merge is a conservative but imperfect correction, and residual misclassification inflates apparent turnover.

Duration is interval-censored and quantized, which is why we use renewal cycles; absolute tenure figures should not be read from this data.

Community area enters both the target definition and the partitioning, so area-level comparisons are circular and excluded from substantive claims.

The single-city scope limits external validity of the urban finding. The evaluation findings depend on the grouped structure of the features rather than on Chicago specifically, but replication across cities remains future work.

Excess turnover is defined relative to a model, so the label inherits that model’s assumptions. Three alternative definitions are reported in the supplement.

8 Conclusion

Spatial blocking and group-aware partitioning address different hazards, and in urban prediction both are usually present at once. When features include aggregates over administrative units, the dominant leakage channel is group membership rather than proximity, and spatial blocks do not close it until they approach the size of those units—well beyond the scale current block-size guidance would suggest. On our task the difference between the default protocol and a correct one

is ROC AUC 0.970 against 0.693, average precision inflated more than sixfold, and a reversal of model selection.

The practical consequence is narrow but, we think, worth stating: a practitioner who identifies their data as spatial, adopts spatial cross-validation and sets block size from predictor autocorrelation has done everything the spatial literature asks and still obtains a fully inflated estimate. The check that would have caught it is cheap—sweep the block size and look for a flat curve—and the fix is to partition by the units the features aggregate over.

The substantive finding survives honest evaluation: excess commercial turnover concentrates at individual addresses rather than districts, with immediate neighbours only weakly elevated, and tenancies at such addresses ending roughly $1.5\times$ faster per renewal cycle.

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Reproducibility

All inputs are public domain: the licence register is published by the City of Chicago as dataset `r5kz-chrr` and may be retrieved by anyone. Section 3 states the extraction date, the row count against which the download was verified, and every threshold used to reconstruct tenancies, so the derived tables can be rebuilt from the published source.

The analysis code, the download utility with row-count verification, and the derived tenancy and location tables are available from the authors on request. The pipeline is deterministic given a fixed seed, which is reported with the code.

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